Porous knee implant variant

Sustaining and design control for a cement-free update

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Why a porous variant?

Knee implants have traditionally been fixed to the bone with bone cement. The porous variant is designed so that no cement is needed during the operation.

Traditional

Fixed to the bone with bone cement during surgery.

Porous

The surface lets bone grow into the implant, so no cement is needed in surgery.

My role on the porous variant

Worked on

Femoral and tibial implants
Porous jigs
Manufacturing file creation

Teams coordinated

CAD team
Manufacturing team
QA team
Process team

Checked and released

Femoral and tibial implants
Porous jigs
Manufacturing files
Drawing files

How the work ran

1. **Design intent**
2. **Design guide requirements**
3. **CAD for implants and jigs**
4. **Manufacturing and drawing files**
5. **Quality checks and release**

Every step ran through the CAD, manufacturing, QA and process teams.

Change control

- Manufacturing requests were handled through controlled DCR releases.
- Corrected documents were issued to the right teams.
- Fixes were delivered in about 2 weeks.

Lesson learned

When one item is fixed, verify and validate the other features in the same DCR before release.

Result

Released without disruption to manufacturing

- Correct files reached manufacturing at release.
- Issues from manufacturing and QA were resolved with the CAD team.

Skills this project shows

Design control	**ECN / ECO / DCR**	**V&V support**
Cross-functional coordination	**SolidWorks**	**JIRA issue tracking**

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